Senthen Velmurugan

Dallas, TX · Open to relocation anywhere in the US (Pacific Northwest / Texas preferred)
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Education

Georgia Institute of Technology **In Progress**
Master of Science in Computer Science (OMSCS) *Atlanta, GA*

The University of Texas at Dallas **Completed**
Bachelor of Science in Computer Science *Dallas, TX*

Experience

KloudGin **May 2024 – August 2024**
Machine Learning Engineering Intern *Remote*

- Raised first-contact resolution 20% over 12 weeks of use by taking the Generative AI team's open question, whether RAG could speed up customer case resolution, from proposal to production rollout
- Gave support engineers suggested next actions from similar past cases and accepted forum solutions by building a multi-source RAG system: an ingestion pipeline, hybrid vector + BM25 retrieval, and cross-encoder re-ranking
- Raised top-k precision by measuring retrieval offline and through an A/B feedback loop with support agents, then diagnosing an overfit to long threads and retuning chunking and metadata filters
- Let other teams integrate it safely with tenant-scoped search, PII scrubbing, safe prompting, and a documented API

GBC Computers **May 2023 – August 2023**
Cybersecurity and Systems Intern *Mabank, TX*

- Improved client network performance by monitoring with PRTG and tracing bottlenecks in Wireshark captures
- Hardened client infrastructure with pfSense firewall rules, Snort intrusion detection, and endpoint protection
- Found and patched critical risks with Nessus vulnerability assessments, validated by OpenVAS and manual audits

Projects

P&ID Digitizer: Scanned Engineering Drawings to Graphs | *Python, PyTorch, OpenCV, FastAPI, Docker* **2026**

- Turned scanned P&IDs into queryable graphs (GraphML, DEXPI 2.0.1 XML) at 0.915 edge F1 and 0.997 symbol mAP@0.5 on a 75-sheet test split by chaining symbol detection, line extraction, and graph assembly
- Found small symbols on 7,000 px sheets by fine-tuning a Faster R-CNN on overlapping 1024^2 tiles, merged across seams
- Sent only doubtful edges to a person by calibrating per-edge confidence for a review UI, shipped as a FastAPI job service in a non-root Docker image with strict mypy and 99 pytest cases in CI

DEXPI P&ID Viewer | *JavaScript, Python, XML, SVG* **2026**

- Let anyone open a DEXPI 2.0.1 plant model with nothing installed or uploaded by building a zero-dependency, single-page viewer that lays out diagrams from topology alone, with no stored coordinates
- Caught invalid files before downstream tools do by validating structure and the metamodel in the browser and a CLI

Word2Eq: Seq2Seq Training & Serving | *Python, PyTorch, FastAPI, Docker, Prometheus* **2024 – In Progress**

- Took valid equations from 97.4% to 100% on 1,000 SVAMP problems by rebuilding a non-functional prototype into a train, evaluate, and serve platform for a 6.4M-parameter Transformer with grammar-constrained decoding
- Served 150 requests/s at 73 ms p95 on one laptop CPU by micro-batching behind a bounded queue that sheds overload with a fast 503, and made training preemption-safe with bit-identical resumes

Campbell Woodworks: Storefront for a Family Woodshop | *Jekyll, Liquid, HTML, CSS, Python* **2026**

- Put a family woodshop's work, once only in Facebook posts behind a login, in front of buyers at no running cost
- Covered 23 pieces with one Markdown file each, and call buttons instead of a cart, since the shop sells by phone

Thai Sizzle: Website for a Family Food Truck | *Jekyll, Liquid, HTML, CSS, Python* **2026**

- Put a Thai food truck's menu, prices, hours, and location online, with all 13 dishes matched to its menu board
- Made it owner-editable with one data file per section, plus a weekly GitHub Action that refreshes the Google rating

Technical Skills

Languages: Python, JavaScript, Go, C/C++, SQL, Bash
Frameworks: PyTorch, FastAPI, LangChain, OpenCV, pandas
Tools: Linux (Debian), Git, Docker, Kubernetes, Terraform, PostgreSQL, Pinecone, CI/CD, Autodesk Fusion

Organizations

Mensa **2013 – Present**